



kokchun giang

how to store non-relational data

Non-relational Data

- Document data are stored as JSON documents.
- Different data entries can have different schemas.
- All contents in an entry can be queried:

Documents - film data example

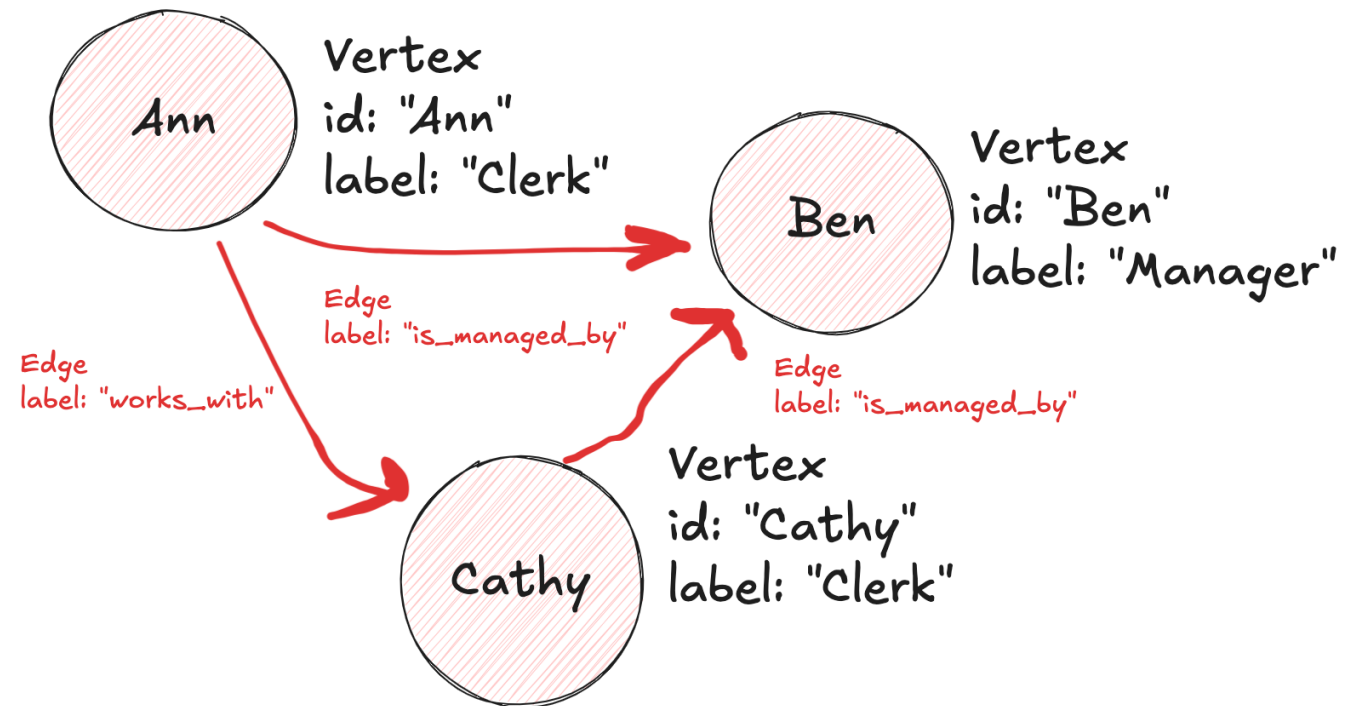
```
{  
  "year" : 2009,  
  "title" : " Tomorrow is another day",  
  "genres" : ["Comedy", "Horror"]  
}
```

```
{  
  "year" : 2025,  
  "title" : "Good luck",  
  "director" : "Kokchun"  
}
```

Non-relational Data

- A graph database stores data entities and relationship between them.
- Data are stored as vertices and edges:

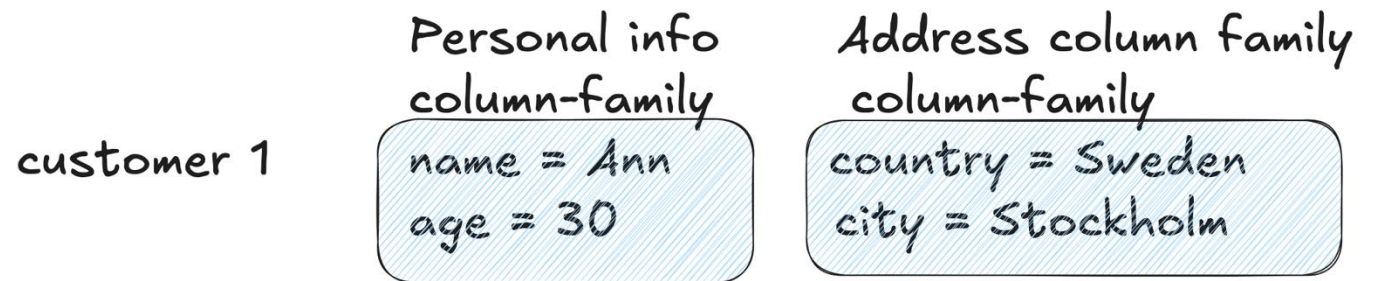
Graph - employment data example



Non-relational Data

- In column-family databases, for each row of record, a subset of column values are stored in one column family.
- queries against a specific column fast and improve performance of analytics application

Column-family data - customer data example



Cosmos DB API

- Under Cosmos DB, we can create instances of Cosmos DB with different APIs for processing different data types:

